

PlayStation 3 Clusters Providing Low-Cost Supercomputing to Universities

Air Force Research Laboratory's Condor Supercomputer begins paying off as researchers publish their results in several fields, including astrophysics.

BY: [Colin Wood](#) | April 25, 2011

Used internally to analyze spy plane imagery and sift through mountains of text, the Condor Supercomputer has been used exclusively by the Air Force Research Laboratory (AFRL). Now, the lab has opened up the supercomputer to universities to conduct research on predictive modeling, computational intelligence, parallel data searches and computational linguistics.

Condor, which was officially announced in December 2010, is said to be among the 40 fastest computers in the world, according to the lab's Condor cluster Project Engineer Mark Barnell. The computer is composed of 1,716 Sony PlayStation 3 (PS3) video game consoles, 168 general purpose GPUs and 78 compute servers powered by 2.67 GHz Intel Xenon X5650 multicore processors.

Neural Networks

Engineers from the University of Dayton in Ohio have been using the Condor cluster to create artificial neural networks. Tarek Taha, an associate professor of electrical and computer engineering at the university, said their research could be a step forward in computing and neuroscience.

"We may not get a walking, talking robot that acts just like us, but at the very least we should be able to get smarter computers," Taha said.

The team has been working primarily with two algorithms, Taha explained. The first uses traditional neural networks to deal with visual problems such as facial recognition and maze navigation. The second algorithm deals with similar problems, but its model is much more "biological," Taha said. It's meant to model the synapses in the brain, exhibiting the same spiking behavior seen in neurons. While the human brain can effortlessly perform tasks such as pattern recognition, language and reasoning, traditional computers are notoriously poor at these things. Structuring the algorithm's neuron models in a more biological way could change all that, Taha said.

Looking forward, Taha predicted that this technology may be suitable for unmanned aerial vehicles. If the algorithm could be implemented in combination with memristors — a nanoscale circuit technology first implemented by Hewlett-Packard in 2008 — then a powerful supercomputer that uses less energy might be designed to fit on board an aircraft. Concerns about data bandwidth to and from the aircraft would no longer be an issue.

The Condor Supercomputer's PlayStation 3, Taha said, is particularly well suited for neural network research. "The PS3's Cell processor handles a lot of parallelism," Taha said, "like neurons in a brain."

When Black Holes Collide

After its release in 2006, Sony's PS3 was seen as an attractive option for high-performance computing. The PlayStation's Cell microprocessor was suitable for intensive computation, the device was Linux compatible and the price was right at around \$500 each. Systems comparable to Condor, built using traditional hardware, can cost 10 times as much. Several research groups developed small PS3 clusters independently soon after the machine's initial release.

One of the first to implement a PS3 cluster was Gaurav Khanna, assistant professor of physics at the University of Massachusetts, Dartmouth. Khanna used a cluster of 16 PlayStation 3s to assist his gravity research, naming the cluster the PS3 Gravity Grid.

Khanna praised Sony's gaming system for its robustness. "They've been running almost continuously for four years now and it's a non-ideal environment. It's a lab, there are students," he said. None of the machines in his cluster have needed replacing, he said.

Khanna is one of the few researchers allowed access to the Condor cluster. "I've used on the order of 300 Playstations so far. That's roughly given me 10 to 20 times more computing power than I'm used to having. That's just huge to me," he said. "I can do runs that are 10 times more accurate or much longer runs. And I'm only using 15 to 20 percent of the machine."

About 100 years ago, Albert Einstein postulated as part of his general theory of relativity that when very massive bodies collide, such as two black holes, they produce ripples in space-time, which he called gravitational waves. No one has yet observed gravitational waves directly, but Khanna hopes his models will reveal what other scientists should be looking for in order to make a direct observation and prove Einstein's postulate. So far, Khanna said, his team's findings mesh with Einstein's theory.

The typical way of detecting phenomena in the universe is light radiation, Khanna explained. The problem with light is that it can be blocked or distorted by other objects in space quite easily. It's not an ideal medium. Gravitational waves, on the other hand, travel very far, Khanna said. "They could come to us. Gravitational waves can travel through miles of lead. They can travel through the earth," he said. If scientists can find a way to detect the waves and harness their power, it might provide a powerful alternative to the way humans have observed the universe for millions of years.

Bragging Rights

In addition to gravity research, Khanna is assisting the AFRL in the development of a benchmarking code that they hope will prove the lab's claim that Condor is among the world's 40 fastest computers. China made news in November 2010 when it took the spot for the world's fastest computer with the record-breaking Tianhe-1A system. According to high-performance computing clearinghouse Top500.org, the U.S. still runs more than half of the world's fastest supercomputers, but that number is dwindling.

While Condor isn't expected to steal back the top spot, the innovative use of consumer electronics in Condor's creation may be a catalyst in the world of high-performance computing. What started as an experiment with just a few hundred devices, quickly impressed the creators of Condor and before long, project engineer Barnell was buying up as many PlayStation 3s as he could find.

The AFRL will release a public paper in June that outlines the results of its research using the Condor cluster.

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